

Raj Kumar Mahto

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PROFESSIONAL SUMMARY

AI Engineer with 6 years of experience engineering production LLM systems, distributed inference, and high-throughput microservices. Architected multi-agent orchestration systems cutting manual provisioning by 80%, built high-scale enterprise RAG pipelines handling 100K+ daily queries over 33M+ records, and enforced automated CI/CD evaluation gates. Deep expertise building low-latency LLM serving platforms with vLLM and Ray, production fine-tuning pipelines (LoRA, SFT, DPO), and cloud-native AI infrastructure on AWS and GCP.

EDUCATION

Arizona State University M.S., Data Science, Analytics and Engineering	May 2026 GPA: 4.0 / 4.0
Visvesvaraya Technological University B.E., Computer Science and Engineering	May 2018 GPA: 3.5 / 4.0

SKILLS

Programming Languages: Python, TypeScript, Go, SQL, Java, JavaScript
Frameworks and Libraries: LangChain, LangGraph, crewAI, Anthropic SDK, OpenAI SDK, Hugging Face Transformers, PyTorch, scikit-learn, FastAPI, Pydantic, Next.js, Node.js, Prisma, Spring Boot, React
Databases: PostgreSQL, pgvector, ChromaDB, Pinecone, Redis, MongoDB
Tools and Software: MLflow, Docker, Kubernetes, Terraform, GitHub Actions, Linux, Kafka, ArgoCD, Ansible, OpenTelemetry, Prometheus, Grafana, SonarQube, Apache Superset, Playwright, tree-sitter, CLI, Docker Compose
Cloud Platforms: AWS (EKS, Lambda, S3, RDS), Azure, GCP

WORK EXPERIENCE

Pebl (Odyn Network - Startup) AI Engineer - Architected an event-driven multi-agent GPU scheduling framework (LangGraph, crewAI, MCP) autonomously routing 80% of inference and fine-tune workloads across multi-cloud GPU providers, saving 15 engineering hours weekly. - Instrumented agent execution loops with LangSmith tracing, OpenTelemetry, and automated failure taxonomies, reducing non-deterministic agent debugging time by 40%. - Fine-tuned a RoBERTa sequence classifier (LoRA, MLflow) predicting request token length before scheduling; dynamically allocated vLLM engine configs and boosted cluster-wide GPU memory utilization from 70% to 95%. - Shipped a self-serve fine-tuning platform supporting SFT and DPO, using VRAM prediction to drive GPU selection and Ray for distributed orchestration; cluster utilization moved from 65% to 98%. - Architected the marketplace control plane in Go, TypeScript, and Next.js across 10+ GPU providers, cutting provisioning time from 12 minutes to under 60 seconds at 99.9% availability. - Optimized high-throughput LLM serving by deploying distributed vLLM instances on 8x NVIDIA A100 nodes via Ray, leveraging Tensor Parallelism (TP=4) and PagedAttention to reduce p99 time-to-first-token (TTFT) by 42%.	Jan 2026 - Present
Infor US LLC AI Engineer - Architected an enterprise RAG microservice on GCP GKS (Python, FastAPI, ChromaDB, pgvector) providing natural-language search over 33M+ ERP transaction records at 100K+ daily queries. - Elevated RAGAS retrieval faithfulness from 0.71 to 0.89 by replacing single-vector search with a hybrid retrieval pipeline (Dense Vector + BM25 Sparse Search) backed by a Cohere Cross-Encoder re-ranker. - Built an automated offline evaluation pipeline integrated into GitHub Actions CI that runs RAGAS scoring (faithfulness, answer relevance, context precision) on LangSmith traces, blocking regression releases automatically. - Engineered a two-tier semantic cache (Redis + in-process LRU) that reduced p50 query latency by 35% and lowered LLM API expenditure by \$12K annually without sacrificing answer groundedness. - Authored automated data-readiness scoring pipelines for pipeline completeness and consistency across 33M records, cutting manual pre-validation efforts by 40%. - Implemented runtime guardrails and PII redaction layers using NeMo Guardrails to sanitize prompts and model responses, preventing prompt injections and maintaining enterprise data privacy compliance.	May 2025 - Dec 2025
Deloitte Software Engineer, AI and Cloud Systems - Built and operated an event-driven microservices platform (Node.js, Kafka, AWS Lambda) synchronizing 10K+ clinical trial records across regulatory systems under HIPAA, sustaining 99.9% uptime. - Designed and secured REST APIs over PostgreSQL, integrating SonarQube quality gates into CI that reduced code complexity 35% and duplication 20%. - Designed a HIPAA-compliant GPT-4 RAG compliance assistant that improved document reviewer throughput by 35% and cut manual effort by 50%. - Migrated legacy BI to Apache Superset dashboards (Python, SQL), eliminating \$15K annual licensing cost and enabling self-serve analytics for 50+ analysts.	Apr 2022 - Aug 2024
Datamatics Software Engineer - Built Node.js microservices for fintech workflows, reducing API latency 40% and automating 90% of client onboarding, accelerating revenue recognition. - Optimized PostgreSQL and Redis caching, reducing response times 65% while sustaining 50K+ daily transactions at peak load. - Delivered PCI-DSS compliant payment integration APIs with zero security incidents during the engagement.	Mar 2021 - Mar 2022
Hexagon Global Associate Software Engineer - Built a modular Report Designer in Java and Spring Boot that cut analyst report creation from 2 hours to 10 minutes. - Optimized data ingestion and API layers for 25% higher backend throughput under sustained load. - Automated CI/CD pipeline with Docker and Kubernetes, reducing release cycle time by 30%.	Aug 2019 - Mar 2021

PROJECTS

Evidence Locker – AI Compliance Evidence Management - Engineered a Next.js 15, PostgreSQL, and Prisma application classifying compliance evidence against NIST SP 800-53 controls using Claude Sonnet with structured outputs. - Utilized prompt caching to slash repeated token costs by 60% and built an async Postgres queue (SELECT FOR UPDATE SKIP LOCKED) to ensure audit trail immutability. - Playwright end-to-end coverage, GitHub Actions CI, and single-command Docker Compose setup.	
Code Quality Intelligence Agent - Built a multi-language codebase analysis agent parsing AST structures with tree-sitter, enabling scope-aware context retrieval beyond standard fixed-window chunking. - Integrated six parallel static analysis engines (security, performance, complexity) orchestrated via FastAPI, CLI, and React UI.	

CERTIFICATIONS & ACHIEVEMENTS

- Agentic AI - DeepLearning.AI
- AWS Certified AI Practitioner (AIF-C01) - AWS
- AWS Certified Developer – Associate (DVA-C02) - AWS
- AWS Cloud Practitioner (CLF-C02) - AWS